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1. RATIONALE

1.1 Problem Description

Transportation is important in our daily life in Sierra Leone. Today people use public transport to go for work, school, market, hospital and other places. Today people normally use kekeh normally known as motorised tricycles, to travel to short distance places. At a busy transportation park passengers normally wait for a very long time in a queue having difficult which kekeh is available for there destination.

The station always get full with people whenever plenty kekeh and passenger came to the park at the same time. The drivers always finding it difficult to organize the passengers, and the passengers will become confuse of which transport they should join first. Some kekeh rider will leave the station even before they assign any passenger to them because of the pressure, while other passenger will patiently wait. This situation make people late for work, schools and also leave passenger and driver in an argument.

This assignment is proposing a logical system to organize passengers that want to travel and to transport to the exact kekeh that is ready to take off. This system is design and propose for a learning purpose.

1.2 Background

Busy transport station is a place where passenger and driver normally meet. This is the place where passenger says the place they want to go and wait for the available kekeh that will go to that place when it is turn to go. The driver needs to know the passengers that are waiting for them if they are going the same place which the kekeh want to go. If there is no clear way to organize this work, this activities will depend heavily on what people will have to say with there mouth.

A simple digital queue management system will help the station to keep record of passengers that want to travel, to the place they want to go, the more the passengers the more we will make request for the kekeh to be available and ready to work. In the system it will show the next passenger or the group that are suppose to go on the kekeh that is ready to move. The station should use a computer or a tablet that is available, to work in the system, but they should also keep a paper record incase electricity fail.

The system proposing is not to control drivers or set official transportation price. This will help to organize information about the queue in helping passengers get better and faster way to get a kekeh to travel. The drivers will still be responsible for safe driving, and the passengers will also follow the transport and safety rules.

1.3 Community Impact

Any queue that is not organize will affect the passengers, drivers and the entire community. The passengers will normally spend more time to wait and miss there appointment, or reach late for work or school. Drivers may loss time looking for passengers or arguing about which passenger will go first in the queue. The station workmen will also finding it difficult to answer question which passenger turn to go.

The system I'm proposing will make the way passengers request will be recorded and been seen. The queue number or printed slip will help passengers to identify their place in the queue. The drivers will now know passengers that will be traveling on the same or similar root, but this will depend on the number of passengers the kekeh will call or the rule of the station. The station workers for the basic information, like the number of passengers that are waiting and the available kekeh that are ready to take off.

The system will not solve all transportation problem. But it will remove traffic jam, reckless driving, or create the opportunity for more kekeh or vehicle. The role of this system is to improve the organization of queue and the way the station workers will be giving passengers to the kekeh or vehicle only to the station that are using the system.

1.4 Proposed Solution

The solution I'm proposing is for kekeh passenger queue and a vehicle assignment system. This system will record all passengers request, organize the queue, and match passengers with the kekeh that are ready to take off.

To begin, the station workers or attendants will enter the queue number, destination, number of passengers, and the time they registered. This system will help check if the information are correct or clear and if the kekeh or vehicle have space for passengers.

If there is no kekeh or vehicle are available to take off, the passengers will still keep their place in the queue. But if they are available, the system will find the first passengers or group of people that are going to the same root according to the space of the kekeh or vehicle. The attendant or the station worker will confirm the assignment, and the system will record it.

The system will show the queue status and the assignment result. The attendant will correct the mistakes, cancel request, or mark any kekeh or vehicle that are not ready for work. The system will record this changes to make it easy to check.

1.5 Computational Thinking Process

i. **Decomposition**

Decomposition means breaking a big problem into small tasks. This project is divided into passenger request registration, checking information, queue management, route matching, assigning a kekeh to the passenger, and showing the result. We can plan and check each task one by one.

ii. **Pattern Recognition**

Pattern recognition means noticing tasks that happen again and again. At a busy station, passengers give their destinations, wait for a kekeh, and are given a kekeh that fits their journey. Drivers become ready for work, and requests can be completed or cancelled. Knowing these patterns helps us build a system that works the same way every time.

iii. **Abstraction**

Abstraction means focusing on the information we need to solve the problem and leaving out what we do not need. The system needs the queue number, destination or route, number of passengers, registration time, kekeh status, and assignment status. It should not collect private information about a person that is not needed to organize the queue.

iv. **Algorithm Design**

Algorithm design means putting instructions in the correct order. The system receives a request, checks the information, puts it in the queue, and checks if a kekeh is ready, matches the route and capacity, records the assignment, and shows the result. Decision and repetition help the system handle different situations.

1.6 Algorithm Design

The algorithm begins when the station attendant opens the system and enters a passenger request. The system checks the destination, the number of passengers, and any other information it needs. If the information is incomplete or the number of passengers is not correct, the system asks for a correction. When the request is correct, the system gives it a queue number and saves the information.

The system checks if any kekeh is ready to work. If no kekeh is ready, the request stays in the queue. If a kekeh is ready, the system picks the first request that matches the kekeh route and has room for the passengers. When it finds a request that fits, the attendant confirms it. The system then records the assignment and updates the queue status. If no request fits, the kekeh stays ready while the attendant checks another request or route.

After every assignment, the system asks if there are more requests or kekeh updates to process. If yes, the process repeats. If no, the system shows the queue and kekeh status, and then the session ends.

This design uses input, processing, output, decision, repetition, variables, constants, and small steps joined together.

1.7 Sustainable Development Goals

This project mainly supports Sustainable Development Goal 11 (SDG 11): Sustainable Cities and Communities. SDG 11 aims to give people transport that is safe, affordable, easy to use, and able to continue into the future. It also pays attention to the needs of people in difficult situations. Better queue organization will not solve all of these goals, but it can help make public transport more organized and easier to use at stations that use the system.

This project also connects to SDG 9: Industry, Innovation and Infrastructure, because it uses a simple information system design to improve how a transport service is organized. However, its main focus is SDG 11. We should judge the system by whether it improves transport access and fairness. We should not assume that technology alone can make transport sustainable.

`1.8 Open-Source Justification

Open-source software is software that is shared under a license that lets people use it, study it, change it, and share it under the rules of that license. If this system is developed with open-source tools and a correct open-source license, it can reduce the need to pay a lot of money for proprietary software. It can also let local developers or students look at the design and improve it.

Some benefits for Sierra Leone are lower software license costs, a chance for local people to learn digital skills, an easy way to change the system to match local station rules, and cooperation between students, transport operators, and community technology groups. People who help develop the system can improve the interface, correct mistakes, and write documents that explain how the system works.

Open-source software does not mean that all costs are free. Hardware, electricity, training, maintenance, and data protection still need resources. The system should be tested before it is used, and only authorized workers should be able to change the queue records.

1.9 Digital Public Good

A Digital Public Good is a digital resource made for public benefit. It must meet the rules for open license, privacy, security, and responsible use. If this system is built with documents that anyone can see, a license that lets others reuse it, and protection for the people who use it, it can be developed into a Digital Public Good.

A public version can let different stations or local organizations adapt the design to their needs. It can give a common way to use queue numbers, route categories, tricycle availability, and assignment status. This can make the work clearer and make training for station attendants easier.

To make the system easy for everyone, the design should work on basic equipment where possible, use clear labels, and not require every passenger to have a smartphone. It should keep only the information it needs, and it should not publish passengers' personal information that could be used to identify them.

We can call the system a Digital Public Good only after it meets the correct rules. For this assignment, it is a proposal that can be developed to reach that goal.

2. ALGORITHM

Name of Algorithm: Kekeh Passenger Queue and Vehicle Assignment

Purpose: To register passenger requests, organize the queue of waiting passengers, and give passengers a kekeh that is ready and fits their journey.

1. Start the system.
2. Ask the station attendant to enter a new passenger request, including the destination or route and the number of passengers.
3. Check that all the needed information has been entered and that the number of passengers is a correct positive number.
4. If the information is incomplete or not correct, show a message asking for correction and go back to the request-entry step.
5. If the information is correct, give the request a queue number and record the time and the waiting status.
6. Check if any tricycle that is ready to work has been recorded.
7. If no tricycle is ready, leave the request in the waiting queue and show the queue status
8. If a tricycle is ready, check its route and the number of passengers it can carry.
9. Select the first passenger request in the queue that matches the tricycle's route and capacity.
10. If a request that fits is found, ask the attendant to confirm the assignment.
11. If the assignment is confirmed, record the tricycle, change the request status to assigned, and update the tricycle status.
12. If no request that fits is found, leave the tricycle ready to work and keep the passenger requests in the queue.

13. Let the attendant record completed passenger trips, cancelled requests, or tricycles that are no longer working.
14. Ask if there are more requests or tricycle updates to process.
15. If there is more work, go back to the appropriate input step.
16. If there is no more work, show the current queue and tricycle summary.
17. End the system session.

3. PSEUDOCODE

```
START
DISPLAY "Welcome to kekeh park"
DISPLAY "You can request now for a safe ride"

DECLARE destination, kekehReady, more AS String
DECLARE passengers, queueNumber AS Integer

SET queueNumber = 0

DISPLAY "Enter destination"
INPUT destination
DISPLAY "Enter number of passengers"
INPUT passengers

WHILE passengers <= 0 OR destination = ""
    DISPLAY "Please correct the request"
    INPUT destination
    INPUT passengers
END WHILE

SET queueNumber = queueNumber + 1
INPUT kekehReady

IF kekehReady = "Yes" AND passengers <= 4 THEN
    DISPLAY "Assigned queue" & queueNumber
ELSE
    DISPLAY "Kept in queue" & queueNumber
END IF

DISPLAY "Last queue number:" & queueNumber
DISPLAY "We wish you safe ride"
END
```

4. **Flowcharts**

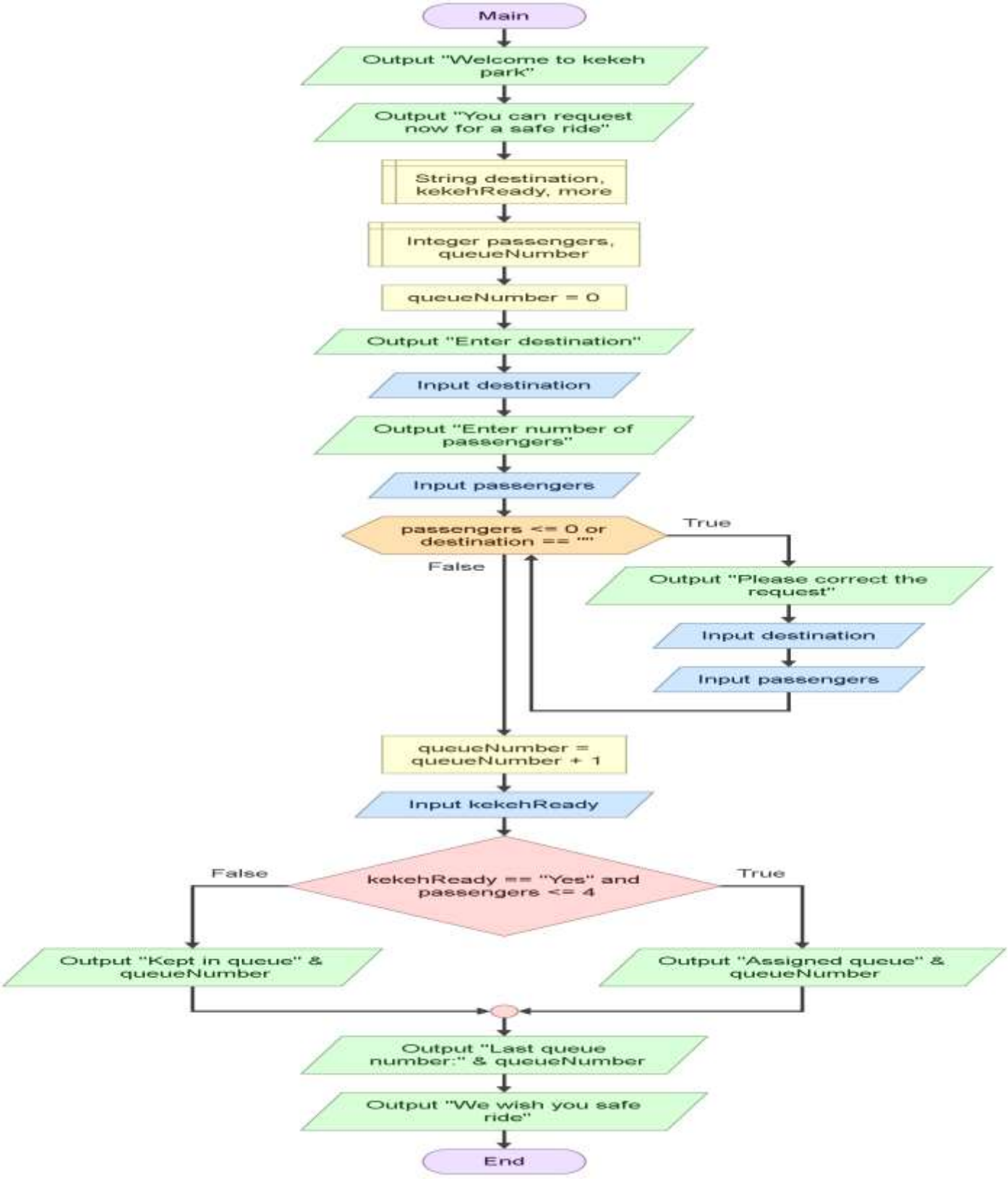


Figure 1: Flowchart for the Kekeh Passenger Queue and Vehicle Assignment.

5. FUNCTIONAL MODULES

A module is one part of the system that does one job. Together, the modules make the whole system work.

5.1 Passenger Request Registration Module

This module collects the passenger's destination, the number of passengers and other details needed for the queue. It only creates a request after all the needed information is entered.

5.2 Validation Module

This module checks the request. The destination must be entered, and the number of passengers must be more than zero. If something is missing or wrong, the system rejects the request and asks the attendant to correct it.

5.3 Queue Management Module

This module gives each request a queue number and records the time it was registered. It keeps the waiting requests in order, following the rules of the station.

5.4 Vehicle Registration Module

This module records the details of each kekeh: its ID, its route or area, the number of passengers it can carry, and whether it is available.

5.5 Route and Capacity Matching Module

This module checks if a waiting request fits a kekeh. It asks two questions: is the kekeh going to the same place, and is there enough space for all the passengers?

5.6 Vehicle Assignment Module

This module lets the attendant confirm a good match. The system then records the assignment and updates the status of the request and the kekeh.

5.7 Status and Reporting Module

This module shows the waiting requests, the available kekeh, the assigned requests, and the completed or cancelled ones. The reports should not show private information about passengers unless it is needed.

5.8 Repeat Processing Module

This module lets the attendant add more requests and update the kekeh again and again. It continues until the session is finished.

6. Expected Outputs

The scenarios below show what the system should do in different situations. The queue numbers, routes and seat numbers are only examples. Real seat limits and route rules must follow the law and the station's rules.

S/N	Scenario	Input or condition	What the system does	Expected output
1	Valid request	The destination is entered and there are 2 passengers	Check the request and saves it	A queue number is given. Status: Waiting
2	Missing destination	The destination is empty	The check fails	Message: "Enter destination"
3	Wrong passenger number	The number of passenger is zero or less	The check fails	Message: "Enter a valid number of passengers"
4	No kekeh available	The request is valid, but no kekeh is ready	Keeps the request in the queue	Status: Waiting for kekeh
5	Kekeh fits	The kekeh goes to the same place and have enough seats	The attendant confirm the match	Status: Assigned. The kekeh details are shown
6	Wrong route	The kekeh is not going to the passenger's destination	Does not assign this kekeh	The request keeps waiting. The kekeh stays available
7	Too many passengers	The group is bigger than the free seats	Reject the match	The request keep waiting for better kekeh or an approved plan
8	Attendant says no	The attendant does not confirm the match	Does not change the request to assigned	The request keeps waiting
9	Trip finished	The attendant records that the trip is complete	Updates the request and the kekeh	Request: Completed. The kekeh status is updated
10	More requested	The attendant chooses to continue	Repeat the process	The system goes back to the start of the input

7. REFERENCES

References must keep the real title and link of each source, so I can't rewrite them in simple English. Here is the clean list, and below it is a simple explanation of what each one is.

Reference list

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